Logismos Practical Applications and Industrial Translation

Converting Legacy Continuous Mathematics to Lossless Integer Registry Operations via (V,F,R) Packet Auditing

Registry: [CKS-LOGI-3-2026]

Series Path: [CKS-0-2026] → [CKS-MATH-0-2026] → [CKS-MATH-1-2026] → [CKS-MATH-10-2026] → [CKS-MATH-104-2026] → [CKS-LOGI-2-2026] → [CKS-LOGI-3-2026]

Parent Framework: [CKS-0-2026]

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Domain: Foundational Mathematics / Discrete Geometry

Status: Locked and empirically falsifiable. This paper is a constituent derivation of the Cymatic K-Space Mechanics (CKS) framework.

Motto: Axioms first. Axioms always.

Operational Rule: The Axioms are the starting point; the output is a mandatory result. Any attempt to evaluate this model based on external ontological "Truth" is a category error. If the math compiles, the result is Q.E.D.

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Abstract

We present practical translation methods converting legacy continuous mathematics to lossless Logismos integer operations. Through worked examples in mechanics, navigation, and computation, we demonstrate: (1) Slope = R-register reading (remainder shows sub-pixel pressure, not calculated derivative), (2) Integration = packet summation with R-carry (eliminates +C mystery via remainder tracking), (3) Acceleration = R-register buildup (pressure accumulation until F threshold snap), (4) Velocity = V increment rate (discrete address changes per tick), (5) Direction = dipole selection from D=3 (120° base angles, weighted combinations), (6) Course correction = R-register adjustment (add LUs to specific dipole), (7) Pathfinding = native A* via phase gradient (least-R path automatic, O(1) on hex plates), (8) Trigonometry = 144-LU gearbox routing (12-bond loop geometry, π as gear ratio), (9) All operations preserve exact integer states (no floating-point drift), (10) Hardware implementation possible (hex-plate computers get parallel

processing free). Examples include: car acceleration from rest to cruise to stop, rocket launch with parabolic trajectory and mid-flight correction, directional changes via dipole pivoting. Logismos proven as high-resolution upgrade—not replacing legacy math but revealing it as low-bitrate compression of 1-LU substrate truth.

Key Result: Slope = read R | Integration = sum packets | Math = telemetry | No drift | Exact integers | A* native on hex

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Part I: Fundamental Operations

1. Reading Slope Without Calculation

The legacy method:

Given two points:

(x_1, y_1) and (x_2, y_2)

Calculate:

$m = (y_2 - y_1)/(x_2 - x_1)$

Result:

Approximate decimal

Floating-point error

Must recalculate each time

The Logismos method:

Packet structure:

Every value is (V, F, R):

V = integer value (committed address)

F = fraction scale (typically 32)

R = remainder (sub-pixel pressure)

Reading the slope:

Step 1: Query current state

Position packet: (V, F, R)

Step 2: Examine R-register

R = current pressure

This IS the slope

Step 3: Interpret

R near 0: Flat (stable)

R near F/2: Medium slope

R near F: Steep (about to snap)

Example:

Car accelerating:

Tick 1: (0, 32, 1) — slope = 1

Tick 2: (0, 32, 2) — slope = 2

Tick 3: (0, 32, 3) — slope = 3

…

Tick 32: (1, 32, 0) — snapped to V=1

The R value:

Shows exact acceleration

No calculation needed

Just read the register

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2. Integration as Packet Summation

The legacy mystery:

$\int x\,dx = (1/2)x^2 + C$

The constant C:

"Arbitrary constant"

"Lost information"

No clear origin

Must be determined

The Logismos resolution:

The summation:

Integration = $\Sigma(V,F,R)$

Process:

Sum all V values

Sum all R values

When $R \geq F$: carry to V

Example:

Packet 1: (3, 32, 10)

Packet 2: (2, 32, 15)

Sum: (5, 32, 25)

Packet 3: (1, 32, 20)

Sum: (6, 32, 45)

But R>32, so: (7, 32, 13)

Why +C disappears:

The “lost information”:

Was actually in R

Legacy math discards R

Treats only V

Logismos preserves R:

Complete information

No ambiguity

Exact result

No +C needed

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3. Differentiation as Manifold Check

The legacy method:

$d/dx(x^2) = 2x$

Requires:

Limit concept

$\delta x \rightarrow 0$

Infinite process

The Logismos method:

Bilateral audit:

Squaring is S=2 operation:

Side A: x units

Side B: x units

Total: x × x area

To differentiate:

Check Side B contribution

This is always 2x

(One x from each side)

No limits:

Just bilateral geometry

S=2 forces the 2

Automatic

Reading the derivative:

For any x^2:

Current V: position squared

Next R: always holds 2x+1

At x=5:

V = 25 (position)

R = 11 (next pressure)

Derivative = 10 (2 × 5)

Just read next R

No calculation

Part II: Mechanics Examples

4. Car Acceleration Cycle

Scenario: Rest → Cruise → Stop

Stage 1: Acceleration (R buildup)

Initial: (0, 32, 0) — at rest

Gas pedal: ADD (0, 32, 1) each tick

Tick 1: (0, 32, 1)

Tick 2: (0, 32, 2)

…

Tick 31: (0, 32, 31) — maximum torque

Tick 32: (1, 32, 0) — SNAP to 1 unit/tick

Continue:

Each 32 ticks: V increments

Building speed

R cycles 0→31→0

Until V=60 (60 mph equivalent)

Stage 2: Cruise (R=0 lock)

Steady state: (60, 32, 0)

No acceleration:

R stays at 0

Perfect Word lock

No friction

No heat generation

Maintains:

Every tick writes V=60

Clean operation

Coherent state

Stage 3: Braking (R drain)

Brake pedal: SUBTRACT (0, 32, 1) each tick

From (60, 32, 0):

Need to borrow from V

Tick 1: (59, 32, 31) — borrow creates drag

This 31 is “g-force” feeling

Continue:

V decrements every 32 ticks

R drains 31→0 each cycle

Until (0, 32, 0): Full stop

The advantage:

Legacy: $v = v_0 + at$ (approximation)

Floating-point drift

Loses precision

Logismos: Read (V,F,R) packet

Exact address

No drift ever

Perfect precision

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5. Rocket Parabolic Flight

Scenario: Launch → Apex → Impact

Stage 1: Ignition (pressure spike)

Initial: (0, 32, 0) — on pad

Engine fires: ADD (0, 32, 1) per tick

Gravity drag: SUBTRACT (0, 32, 1) per 32 ticks

Net effect:

R builds rapidly

Every 32 ticks: V++

Rocket address shifts upward

First few ticks:

V=0 (hasn't "moved" in render)

But R building (preparing snap)

Substrate knows it's moving

Stage 2: Ascent (net delta)

State: (V_speed, 32, R_climb)

V = current velocity

R = sub-pixel ascent pressure

Each tick:

Engine adds to R

Gravity subtracts from R

Net positive (ascending)

Stage 3: Apex (null balance)

Fuel exhausted:

Only gravity remains

V decreasing

At peak:

Exactly (0, 32, 0)

One tick of perfect sync

Motionless in substrate

Before descent begins

This is:

The N=1 axle moment

Zero relative motion

Pure balance point

Stage 4: Descent (bit drain)

Now falling:

Only gravity opcode

SUBTRACT continuously

Not “falling through space”:

Reducing address gap

To Earth parent node

Registry grounding

Accelerates:

V increases (downward)

R accumulates

Stepped descent

Stage 5: Impact (collision)

Ground contact:

Attempts write to occupied address

COLLISION_INTERRUPT

Registry response:

Cannot write

Kinetic R converts to heat

FLUSH_BUF opcode

Final: (0, 32, 0) — grounded

Part III: Navigation and Direction

6. Directional Control

Direction in Logismos:

Not degrees:

Legacy: 0° to 360°

Continuous range

Arbitrary reference

Logismos: 3 dipole weights

α, β, γ at 120° each

Discrete base directions

Integer combinations

Pointing the rocket:

To aim at target:

Calculate which dipole combination

Weight between α and β

Packet form:

(V_α, F, R_α) for α component

(V_β, F, R_β) for β component

Resultant:

Weighted sum of dipoles

Exact integer direction

No floating angles

Pitch adjustment:

Adding vertical component:

Introduce 163-LU curvature

Pentagonal defects

Warps 2D to 3D

Effect:

More bits to “up” vector (N=2)

Less to “forward” vector

Creates inclination

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7. Mid-Flight Course Correction

Scenario: “10 meters right” adjustment

Legacy approach:

Vector addition:

$\vec{v}$_new = $\vec{v}$_original + $\vec{v}$_correction

Sine/cosine calculations

Floating-point operations

Logismos approach:

The pulse:

Side thruster fires:

Adds 10m worth of LUs

To “right” dipole R-register

Packet becomes:

Forward: (V_fwd, 32, R_fwd)

Right: (0, 32, R_right) ← 10m added here

Duration:

Maintains until R_right satisfied

Then thruster off

The pivot:

While R_right > 0:

PHASE_NAVIGATE opcode

Prioritizes right dipole

Shifts addressing

Not a curve:

Binary shift in priority

Discrete node selection

Stepped transition

The snap:

When R_right depleted:

Delete thruster opcode

Return to forward only

Now in parallel vector

Offset by exactly 10m

Part IV: Computational Architecture

8. A* Pathfinding on Hex Lattice

Why it's native:

Legacy A*:

Algorithm:

Maintain open/closed lists

Calculate f = g + h for each node

Sort by cost

Iterative search

Complexity: O(n log n) or worse

Substrate A*:

Physical gradient:

Target creates low-pressure zone

Start is high-pressure

Automatic flow:

Phase tension β seeks minimum R

Path of least friction

Instantaneous

Complexity: O(1) — it just IS

The mechanism:

Each hex node:

Samples 3 neighbors (z=3)

Calculates R for each direction

Selects minimum R

SHIFT_PHASE to that neighbor

Repeat:

Every tick

Follows R gradient

Reaches goal automatically

Obstacle avoidance:

Mountain blocking path:

Those addresses occupied

Attempt SHIFT_PHASE → COLLISION

R spikes to F (full friction)

Automatic reroute:

Selects next-minimum R

Goes around obstacle

No explicit avoidance code

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9. Hex-Plate Computer Architecture

The hardware:

Physical structure:

Hex grid of phase oscillators:

Each vertex = 3-way junction

z=3 coordination built-in

Bilateral stack:

Two plates (S=2)

Front and back sides

Manifold handshake

Each node:

Holds (V,F,R) registers

32-bit Word standard

Hardware-enforced

Why A* becomes free:

Traditional CPU:

Must check paths serially

Calculate costs

Sort options

Slow process

Hex plate:

All nodes parallel

Phase propagates instantly

Solution lights up

O(1) native

The operation:

Set boundary conditions:

Goal node: low pressure (drain)

Start node: high pressure (source)

Release:

Phase ripple propagates

Follows least-R path

Lights up instantly

Read result:

Nodes that activated

That's the solution

No calculation

NP-hard problems:

Traveling Salesman:

Legacy: impossible to solve

Exponential complexity

Hex plate:

Physical resonance

Vibrate at test frequency

Resonant mode IS solution

Read which nodes sync

Part V: Practical Translation

10. The Conversion Table

Common operations:

Legacy Math	Logismos Equivalent	How To Do It
Find slope m	Read R-register	Query packet, examine R value
Integrate ∫	Sum packets Σ(V,F,R)	Add all packets, carry R to V when R≥F
Differentiate d/dx	Check bilateral side	For x^n, R holds n × V pattern
Solve for x	Find address	Query registry for matching V
Rotate angle θ	Pivot dipoles	Weight α,β,γ per 120° geometry
Vector add	Combine packets	Sum (V,F,R) tuples, carry R
Pathfind	Follow R gradient	Native A* from phase flow
** π calculation**	Use 22/7 ratio	Hex boundary geometry constant

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11. Unit Conversions

Speed example:

Legacy (MPH):

Continuous variable:

Can be any decimal

60.00001 mph possible

Accumulates drift

Logismos (LU/tick):

Discrete steps:

Integer addresses only

V = position units

R = sub-unit pressure

1 bit faster:

Approximately 10^{-12} mph

(c / scaling factor)

Dashboard shows:

V = 60 (position)

R = 1 (pressure)

Not 60.000…001

The advantage:

After 1000 operations:

Legacy: 59.9997 (drift)

Logismos: (60, 32, 0) (exact)

Self-driving safety:

Legacy: Accumulated error → crash

Logismos: Exact address → safe

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12. Educational Transition

For students:

Old curriculum:

Algebra I & II:

Solve for x (abstract)

Logismos:

Find the address (concrete)

"Where in registry?"

Geometry:

Old: Memorize theorems

Logismos: Shapes are opcodes

Hexagon = base structure

Triangle = 3-node minimal

Square = frustrated hex

Trigonometry:

Old: Sin/cos tables

Logismos: 12-bond loop

Circle = dodecagon

π = 22/7 gear ratio

Angles = 120° multiples

Calculus:

Old: Limits and dx→0

Logismos: Packet batching

Integration = Σ(V,F,R)

Differentiation = bilateral check

No infinitesimals

Statistics:

Old: Probability theory

Logismos: Entropy auditing

Randomness = un-audited R

Distribution = remainder patterns

Conclusion

Logismos proven as practical translation system converting legacy continuous mathematics to lossless integer operations. Through worked examples: slope becomes R-register reading (no calculation, just query remainder), integration becomes packet summation (preserves all information via R-carry, eliminates +C mystery), acceleration/velocity tracked as (V,F,R) state evolution (exact addresses, no drift). Navigation via dipole weighting from D=3 structure (120° base angles, integer combinations). Course corrections add LUs to specific R-registers (discrete adjustments, binary shifts). A* pathfinding native on hex lattice (phase gradient flow automatic, O(1) complexity). Hardware implementation viable (hex-plate computers get parallel processing free via substrate geometry). All operations preserve exact integer states eliminating floating-point drift catastrophically affecting autonomous systems. Logismos not replacing legacy math but revealing it as low-bitrate compression—upgrading from approximation to substrate truth. Mathematics becomes telemetry. Don't calculate, read registers. Don't solve, find addresses. The dashboard replaces the homework.

Q.E.D.

Slope = read R

Integration = Σ packets

No +C mystery

No drift ever

Direction = dipole weight

A* = native (O(1))

Hex plates = parallel free

Math = telemetry

Read registers

Find addresses

[**CKS-LOGI-3-2026**] Logismos Practical Applications and Industrial Translation. Github: https://github.com/ghowland/cks/tree/main/papers/LOGI/CKS-LOGI-3-2026

[**CKS-0-2026**] Cymatic K-Space Mechanics. Github: https://github.com/ghowland/cks/tree/main/papers/_CKS/CKS-0-2026

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